UNIVERSITY OF NUEVA CACERES
School of Computer and Information Sciences
Bachelor of Science in Information Technology • Naga City, Camarines Sur

TIME-MOTION AND WORKFLOW LATENCY LOG SHEET

Step 4: Objective Timing Instrument for Baseline (Pre) vs. System (Post) Processing

RESEARCH TIMELINE STAGE: STEP 4 (TIMING OBSERVATION - BASELINE & POST-TEST)

When to Use: Used in two phases: (1) *Baseline Run* before system deployment to record current manual/eRPT delays, and (2) *Post-Test Run* to record turnaround times using the web synchronization system.
Primary Quantitative Variables: (1) *Document Processing Latency* (days/hours) and (2) *Record Reconciliation Time* (minutes per property).
Instructions: Log timestamps for each processing phase per case code (e.g., TOA-001). Do not record personal taxpayer details.

Case Transaction Code: _____	**Run Condition:** [ ] BASELINE (Manual/Current) [ ] SYSTEM (Proposed)
Transaction Type: [ ] Transfer of Assessment [ ] New Tax Declaration / Building [ ] Reassessment / Value Revision [ ] Tax Clearance / Certification [ ] Other: _____	**Observed Department(s):** [ ] Municipal Assessor's Office [ ] Municipal Treasurer's Office [ ] Provincial Assessor's Office (Pili) **Data Classification:** [ ] Masked / Dummy Test Data [ ] Actual Workflow Log
Date of Observation: _____	**Lead Observer / Researcher:** _____

TABLE A: PHASE-BY-PHASE TRANSACTION PROCESSING LATENCY

Ph	Operational Process Phase	Dept	Start Timestamp (Date / HH:MM)	End Timestamp (Date / HH:MM)	Duration (Min / Days)	Bottleneck / Error Remarks
1	Counter intake & initial documentary completeness verification	MAO				
2	Property record search, manual ledger/eRPT parcel retrieval	MAO				
3	Field appraisal calculation, valuation encoding, FAAS prep	MAO				
4	Municipal Assessor review, endorsement & signature	MAO				
5	Inter-office document transfer to Municipal Treasurer queue	Transit				
6	Treasurer tax payment cross-check & clearance validation	MTO				
7	Physical/digital batch transport to Provincial Assessor (Pili)	Transit				
8	Provincial appraisal evaluation, title review & formal approval	PAO				
9	Return transit, final municipal encoding & Tax Declaration release	MAO				

TOTAL ELAPSED TIME (Intake to Release): _____ Days _____ Hours _____ Minutes
NET ACTIVE WORK TIME (Total Processing): _____ Minutes / Hours
QUEUE / INTER-OFFICE TRANSIT LATENCY: _____ Days / Hours

TABLE B: RECORD RECONCILIATION AND PAYMENT VERIFICATION METRICS

Batch Reconciliation Parameter	Observed Metric Value	Operational Context & Methodological Notes
Total Property Records Checked in Batch	_____ records	Sample batch of property accounts cross-referenced against collection rolls.
Batch Verification Start Timestamp	Date: _____ Time: _____	Clock time upon initiation of record cross-checking procedure.
Batch Verification Completion Timestamp	Date: _____ Time: _____	Clock time upon resolution/verification of the final record in batch.
Total Batch Execution Duration	_____ minutes	Net active minutes elapsed to reconcile the complete record batch.
Average Reconciliation Latency per Record	_____ min/record	Average active minutes elapsed per individual record verification.
Discrepancies / Data Mismatches Detected	_____ mismatches	Frequency of PIN, owner, or assessment valuation mismatches identified.

Reconciliation Mechanism Employed	[ ] Manual Ledgers / Excel [ ] Proposed Sync Engine	Comparative verification method utilized during observation.

TABLE C: COMPARATIVE EVALUATION SUMMARY AND REDUCTION ANALYSIS

Performance Indicator Metric	Baseline Mean (T_0)	System Mean (T_1)	Absolute Difference (Δ)	Percentage Reduction (% Red)
Document Processing Latency (Transfer of Assessment, days)	_____ days	_____ days	_____ days	_____ %
Building Assessment End-to-End Latency (days)	_____ days	_____ days	_____ days	_____ %
Record Reconciliation Latency per Property (minutes)	_____ min	_____ min	_____ min	_____ %
Frequency of In-Person Status Inquiries per Transaction	_____ visits	_____ visits	_____ visits	_____ %

OBSERVER VERIFICATION & SUBMISSION

I hereby certify that the timestamps and duration metrics recorded in this observation log accurately reflect the observed workflow times.

Observer Signature: _____ Verification Date: _____
Printed Name: _____ Office / Station: _____